

AI LAB 2.py - C:/Users/K Aishwarya/Documents/AI LAB 2.py (3.14.2)

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```
def is_safe(board, row, col):
    for i in range(row):
        if board[i] == col:
            return False

        if abs(board[i] - col) == abs(i - row):
            return False

    return True

def solve_queens(board, row):
    if row == 8:
        return True

    for col in range(8):
        if is_safe(board, row, col):
            board[row] = col

            if solve_queens(board, row + 1):
                return True

            board[row] = -1

    return False

board = [-1] * 8

if solve_queens(board, 0):
    print("Solution:")
    for row in range(8):
        for col in range(8):
            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
else:
    print("No solution")
|
```

>>>

===== RESTART: C:/Users/K Aishwarya/Documents/AI LAB 2.py =====

Solution:

```
Q . . . . .
. . . . Q . .
. . . . . Q
. . . . Q . .
. . Q . . . .
. . . . . Q .
. Q . . . . .
. . . Q . . .
```

>>>